MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	CSM-C12

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Title of the Project: **AI-Powered Incident Management Agent**

Organizations receive incidents through multiple channels such as web portals, email, APIs, chat interfaces, monitoring systems, and operational alerts. These incidents may involve application failures, server issues, network problems, database errors, or cloud infrastructure disruptions. Traditional incident management systems rely heavily on manual classification, rule-based workflows, and human intervention to understand incident descriptions, classify incidents, determine priority, identify root causes, assign tickets to appropriate support teams, and recommend resolution steps. Information required for incident handling is distributed across knowledge bases, monitoring systems, log stores, configuration databases , and ticketing systems, making the overall process time-consuming and difficult to manage efficiently. Existing ticketing platforms provide workflow automation but do not automatically understand incident reports, retrieve relevant Standard Operating Procedures and previous incidents, analyze monitoring and log data together, identify possible root causes, recommend resolutions, perform automated remediation for suitable incidents, or continuously learn from resolved incidents.

This project proposes an AI-Powered Incident Management Agent that integrates a Large Language Model (LLM), knowledge bases, monitoring systems, configuration databases, ticketing platforms, and log analysis into a unified incident management framework.The workflow begins when a user, employee, or monitoring system reports an incident through a portal, email, API, or chat interface. The request is sent to the API Gateway, which forwards it to the Incident Management Agent (LLM). The agent reads the incident description, identifies the type of problem, predicts its priority, analyzes possible causes, and prepares a summary. It then collects relevant information from the Knowledge Base (RAG), monitoring systems, log stores, and the CMDB to understand the incident more accurately. Using this information, the AI Decision Engine finds similar past incidents and recommends the best solution. If the AI is highly confident, it can automatically execute a predefined auto-healing script; otherwise, it assigns the incident to the appropriate support team and sends notifications through email or communication tools. After the incident is resolved, the solution is stored in the Knowledge Base, creating a learning feedback loop that helps the system handle similar incidents more efficiently in the future.

The proposed system aims to improve incident response efficiency, reduce manual effort, enhance decision consistency, support automated remediation, enable continuous learning from resolved incidents, and reduce Mean Time to Resolution (MTTR) through intelligent incident triage, automated diagnosis, and AI-assisted resolution within a unified incident management framework.

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Project Guide

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